SE – LAB -03

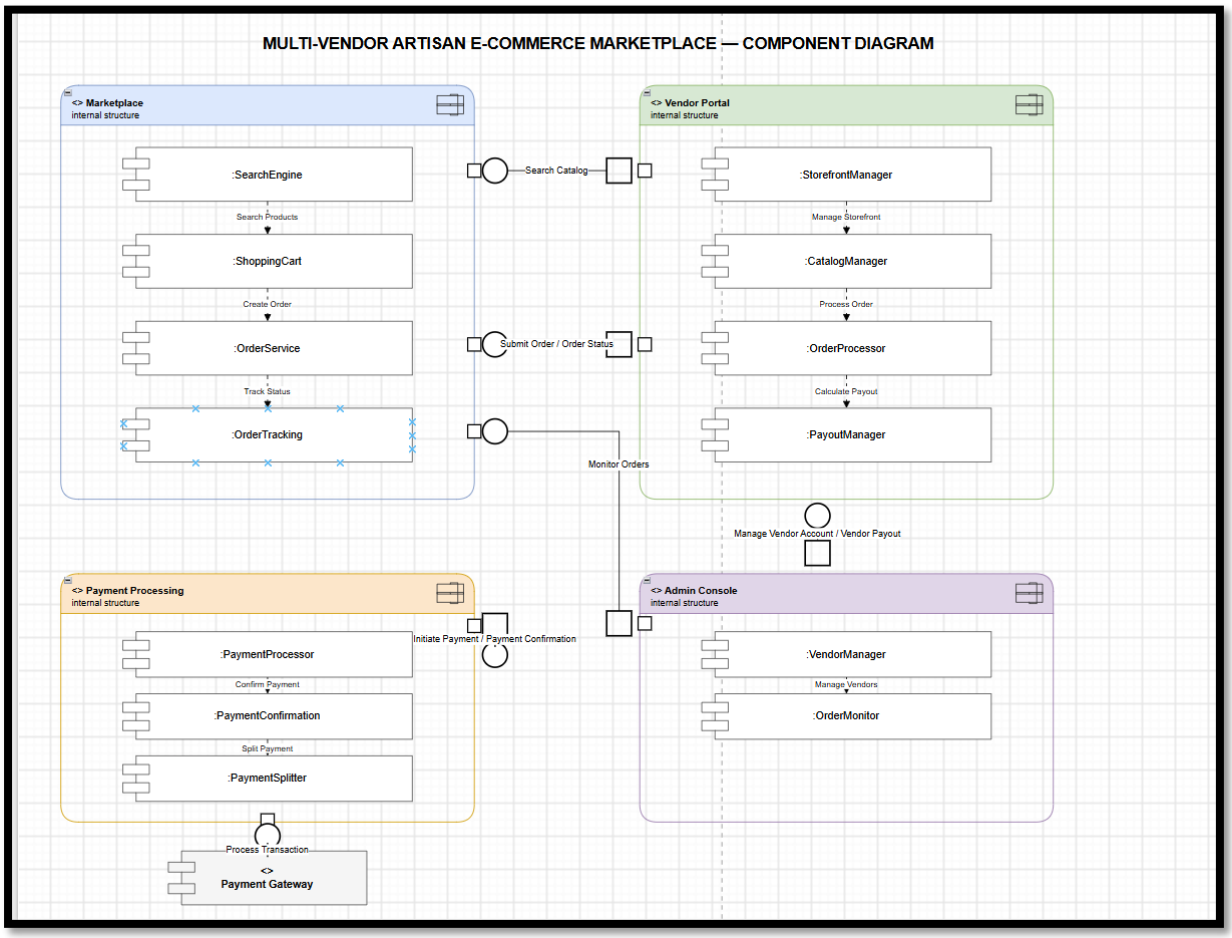
COMPONENT DIAGRAM – MULTI – VENDOR ARTISAN E=COMMMERCE MARKETPLACE

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COMPONENT DIAGRAM:



EXPLANATION:

The component diagram is justified by dividing the system into separate subsystems based on their responsibilities. The Marketplace subsystem handles customer related activities such as searching products, managing the shopping cart, placing orders, and tracking orders. The Vendor Portal is separated because vendor activities such as managing storefronts, catalogs, orders, and payouts are different from customer activities.

The Payment Processing subsystem is kept separate because payment related operations require independent processing and interaction with an external Payment Gateway. This separation also improves security and makes the payment functionality easier to maintain.

The Admin Console is separated because administrative operations such as managing vendors and monitoring orders are different from normal customer and vendor operations.

The components within each subsystem are organized according to their individual responsibilities. For example, OrderService handles order related operations while OrderTracking handles tracking separately. This provides modularity and reduces dependency between unrelated functionalities.

The lollipop and socket interfaces are used to show the services provided and required between components. This makes the communication between subsystems clear. The dashed dependency connections show how one component depends on another to complete a particular operation.

Overall, the architecture provides separation of responsibilities, modularity, maintainability, scalability, and clear communication between different parts of the system while covering the major functional requirements of the e commerce marketplace.